ALOK SINGH KATARIA

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PHYSICS · COMPUTATIONAL SCIENCE · SOFTWARE ENGINEERING · AI · QUANTUM COMPUTING

Objective

Seeking a role in **academic research, EdTech, or academic software development** where I can combine my foundation in **Physics, Mathematics, and Numerical Methods** with hands-on software engineering and DevOps experience. Specifically interested in opportunities involving **computational science, scientific software, educational platforms, or research tooling** — where rigorous mathematical thinking, numerical simulation, and modern software practices intersect to solve meaningful problems in academia or education.

Profile

Physicist and Senior Software Engineer with 25+ years of professional experience across software development, cloud platforms, DevOps, automation, enterprise systems, and EdTech. **IIT Kanpur Physics graduate** with a strong foundation in quantum mechanics, mathematical physics, linear algebra, probability, and numerical methods.

Hands-on experience with Python, scientific computing, AI/ML, and quantum-computing tools. Independent technical work includes **quantum mechanics, numerical methods, Hamiltonian and eigenvalue problems, VQE, quantum simulation, quantum chemistry, and quantum optimization** using Python and Qiskit.

Combines Physics and mathematical foundations with extensive software-engineering experience to **model, simulate, analyze, and prototype computational solutions**. Also experienced in developing mathematical and educational software and working with AI/ML approaches in EdTech.

Interested in opportunities involving Physics, computational science, scientific software, EdTech, AI/ML, and practical applications of quantum computing.

Core Areas

`Computational Physics` · `Quantum Mechanics` · `Mathematical Physics`
`Scientific Computing` · `Numerical Methods` · `Linear Algebra` · `Optimization`
`Quantum Computing` · `Quantum Algorithms` · `Quantum Simulation`
`VQE` · `Quantum Chemistry` · `Quantum Optimization`
`Quantum Noise & Error Mitigation`

Selected Quantum & Computational Physics Work

- Applied classical numerical methods to quantum systems including the **hydrogen atom, 1-D particle-in-a-box, and harmonic oscillator**, using Python, NumPy, SciPy, linear algebra, and the **Finite Difference Method (FDM)** to construct Hamiltonians and compute energy eigenvalues, comparing numerical results with analytical solutions where available.
- For systems without convenient closed-form solutions, explored approximate numerical approaches using Python and **AI-assisted computational methods** for formulation, implementation, and validation.
- Applied a similar computational approach to quantum-computing formulations, using **qubit representations, Hamiltonians, quantum circuits, and VQE** with Python and Qiskit.
- Explored quantum-chemistry workflows using **STO-3G, Hartree–Fock, molecular Hamiltonians, and variational ansätze** for ground-state energy estimation, comparing results with classical calculations.
- Used classical numerical solutions as reference points for exploring corresponding **quantum formulations and variational approaches**, along with quantum optimization and quantum-information concepts such as portfolio optimization, qubit encoding, quantum randomness, spin dynamics, Hadamard transformations, and measurement probabilities.

Professional Experience

IBM — Advisory Software Engineer | DevOps / SRE | 2022 – Present

1. CSUtils Ecosystem — Platform Architecture, Resilience & L3 Engineering

- Led architecture modernization and resilience engineering across IBM Cloud security utilities, modernizing Kubernetes configuration with CRDs, strengthening VPN connectivity with fail-fast validation, and ensuring reliable syslog delivery across pod restarts.
- Designed automated multi-cluster chaos testing and owned L3 production troubleshooting across Kubernetes, VPN, TCP/UDP connectivity, and networking failures to improve service reliability.

2. Dreadnought — IBM Cloud PaaS Re-platforming & Cloud Engineering

- Drove IaC-based provisioning and re-platforming of IBM Cloud infrastructure, including VPC networking, OpenShift, IAM, managed databases, and CI/CD integrations using standardized platform configurations.
- Implemented private endpoint connectivity, PAG/GPVPN integration, bastion access, and security scanning; supported complex cloud networking and platform escalations through root-cause analysis and remediation.

3. Compliance Automation — AI Agent, Auditree & CI/CD Engineering

- Led Auditree onboarding and Python-based compliance automation for SOC 2-aligned evidence collection, security posture assessment, centralized evidence storage in IBM Cloud COS, and Git-based remediation workflows.
- Initiated an AI agent for troubleshooting Jenkins and Auditree pipelines, leveraging repository documentation for failure analysis and remediation recommendations; integrated automated workflows with CI/CD practices spanning Jenkins and Tekton.

Wipro — Lead Software Developer | 2022

Software development for the **CreditMemo** process of the Commissioning System for **Telefónica Germany**, building Python/Flask/Pandas services for data loading, aggregation, and validation.

- Implemented feature enhancements, defect fixes, and unit testing; worked with PostgreSQL and Google Cloud.

RevenueBox — Senior Software Engineer | 2021

B2B Account-Based Marketing (ABM) and go-to-market software focused on identifying, prioritizing, and engaging the right customer accounts.

- Developed a Python microservice to identify, process, and select relevant customer accounts based on multiple input signals.
- Designed a mathematical scoring model to calculate and assign a **"Quibble Score"** for account prioritization and tracking, using Python, Docker, MongoDB, and Google Cloud.

Lumos Learning — Academician / Educator & Research Scientist | 2019 – 2020

EdTech platform delivering dynamic, animated, and guided-practice learning experiences for **K-12 students**, aligned to Common Core Standards (corestandards.org), with emphasis on mathematics, automated question generation, and illustrative scientific concepts.

Drove the platform's modernization end to end — uniquely bringing **Physics, Mathematics, animation-driven software development, and serverless cloud architecture** together in one frame — and led the addition of several new modules spanning dynamic question creation, guided practice, and a mathematics chatbot.

- Led development of **Python-based animated and interactive question-generation modules** that expose K-12 students to dynamic, visually engaging problems with **varied difficulty levels** — catering to the full ability spectrum, from remedial to advanced.
- Architected and built **guided-practice workflows** with automated solution processing, difficulty estimation, and adaptive evaluation, enabling step-by-step problem solving aligned to Common Core standards.
- Led the design of a **mathematics-focused chatbot** and educational automation initiatives, applying NLTK-based natural language processing and classical neural-network/machine-learning techniques of the period.
- Designed the **serverless cloud architecture** using Python, AWS Lambda, and SQL Server to support these modules.
- Illustrated advanced mathematical and scientific concepts — including **solving quantum Hamiltonians via the Finite Difference Method and Fourier Transform techniques** — in simplified, visual, and accessible ways suitable for curious students and educators.

HCL Technologies — Senior Software Engineer | 2017 – 2018

L3 product support for **IBM AppScan** (web application security testing), troubleshooting complex customer issues, root-cause analysis, and coordinating fixes across SAST/DAST scanning and reporting workflows.

- Worked on AppScan SAST/DAST capabilities, customer configurations, defect analysis, and production issue resolution.

IBM India Software Labs — Senior Staff Software Engineer | 2008 – 2017

L3 Development & Troubleshooting Engineer for **IBM InfoSphere Data Replication (IIDR)** (previously DataMirror), specializing in complex production issues, root-cause analysis, debugging, and customer escalations.

- Expert in heterogeneous data replication and integration, including **DB2 for i / AS400, Oracle, DB2, SQL Server, and Teradata**, with IBM messaging and remote journaling technologies.
- Strong experience in cross-language and cross-platform integration, including **AS400/IBM i** and COBOL-based enterprise systems; supported customer ETL and data-movement workflows.
- Recognized as a performance specialist, contributing to performance analysis and optimization and receiving an **IBM award for performance expertise**.

- Contributed to feature development, product maintenance, automation, and build/release pipeline engineering across multiple product releases.
- Worked with C, Java, Python, RPG, COBOL, AS400, and Jenkins.

CA Technologies — Software Engineer | 2005 – 2008

Worked on **eTrust Access Control**, **eTrust Audit**, and **eTrust Single Sign-On** — enterprise security products.

- Developed more than 100 automated tests for nightly builds and ported product code to AIX.
- Investigated customer and QA defects including memory corruption, enhancements, and installation/upgrade issues.
- Worked with C, Python, Linux, UNIX, AIX, Solaris, and HP-UX.

Technical Skills

Programming: Python, C, Java, Shell, JavaScript

Scientific Computing: NumPy, SciPy, Pandas, MATLAB, scikit-learn, TensorFlow

Quantum Computing: Qiskit, VQE, Quantum Circuits, Quantum Simulation

Cloud & DevOps: IBM Cloud, AWS, Google Cloud, Kubernetes, OpenShift, Docker, Jenkins, Tekton, CI/CD

Databases: PostgreSQL, MongoDB, SQL Server, Oracle, DB2, Teradata

Operating Systems: Linux, AIX, Solaris, HP-UX, Windows, IBM i / AS400

Selected Research & Technical Work

Quantum Intuition: From Spin Dynamics to the Hadamard Gate and π Estimation via Quantum Randomness

Explored quantum randomness and quantum-state transformations through Qiskit-based simulations, including spin dynamics, Hadamard transformations, measurement probabilities, and Monte Carlo estimation of π.

Status: Under IBM internal review for publication.

Classical VQE Demonstration for the Hydrogen Atom

Developed a classical demonstration of the VQE approach for the hydrogen atom, connecting the variational principle, Hamiltonian formulation, parameterized states, energy expectation values, and numerical optimization.

Status: Under IBM internal review for publication.

Intellectual Property

- **US Patent #9,528,562** — Granted US patent related to mechanics.
- Two additional mechanics-related inventions under IBM patent review.
- Internal IBM intellectual-property work related to **quantum portfolio optimization**.

Education

IIT Kanpur — Integrated M.Sc. in Physics | 2000

Academic foundation includes Quantum Mechanics, Classical Mechanics, Electromagnetism, Mathematical Physics, Linear Algebra, Probability, Differential Equations, Numerical Methods, and Statistical Physics.

GATE — Physics | AIR 29 | 2003

Academic Distinctions & Awards

- **IIT-JEE AIR 1211**
- **IBM Excellence and Eminence Award** — 2012
- IIT Kanpur alumnus
- Strong academic foundation in Physics and Advanced Mathematics

Physics Teaching & Mentoring

IIT-JEE Advanced Physics Faculty / Mentor

Narayana · PIE Education · Vizag Vikas College

- Taught advanced IIT-JEE Physics with emphasis on **conceptual understanding, mathematical formulation, and rigorous problem solving**.
- Worked extensively with mechanics, electromagnetism, modern physics, and mathematical approaches to advanced Physics problems.

- Mentored students preparing for highly competitive engineering entrance examinations.

Research & Code

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Selected work includes computational quantum mechanics, numerical Hamiltonian methods, quantum simulation, VQE, quantum chemistry, quantum optimization, and quantum-information experiments using Python and Qiskit.